4orm FINANCE

Future-Level Hazard Assessment

Applying preventive safety principles and counselling-centred AI to major financial decisions.

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4orm Finance · Confidential
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CONTENTS

Six parts, forty-two sections and thirty-nine exhibits. Every figure carries its source.

Executive summary	Prevention moved risk discovery upstream; finance can test the same discipline.	4
Introduction	The challenge posed, and why the founder asked it.	6
Part 1 · High-risk industries learned to prevent harm	From reacting to incidents to finding hazards before work begins.	8
Part 2 · Finance has a pre-decision risk problem	Context, future risk and evidence can arrive too late.	19
Part 3 · What counselling-centred systems can teach finance	What disclosure, coaching and conversational research teach finance.	27
Part 4 · The Future-Level Hazard Assessment	A proposed model for finding and controlling foreseeable hazards.	38
Part 5 · Private first. Shared by choice	Private exploration, permissioned data and a better professional handoff.	46
Part 6 · From hypothesis to evidence	Pilot design, measures and kill tests for the thesis.	52
Conclusion	What the evidence supports, and the restrained thesis that follows.	56
Appendix · Method and evidence grading	How the evidence was weighed, and the commercial disclosure.	57
Sources	Forty-one references, each linked from the page that cites it.	59

This paper uses “Counselling-Centred AI” and “Future-Level Hazard Assessment (FLHA)” as proposed 4orm concepts. They are not established academic or regulatory categories. A reference in square brackets is a link to its source entry at the back.

The forty-two sections.

01	Safety improved when risk discovery moved upstream	9
02	Field-level assessment shows the risk before work begins	10
03	Likelihood × severity turns concern into a decision	11
04	Controls matter more than warnings	12
05	Process safety widened the lens from the worker to the system	13
06	Prevention through Design moves protection earlier	14
07	Frontline safety is continuous, not one-and-done	15
08	PCL shows prevention operating at scale	16
09	Syncrude and Canadian Natural make frontline control explicit	17
10	The lesson is prevention, not equal consequences	18
11	The financial decision starts before the application	20
12	Regulation expects understanding. Execution is uneven	21
13	The missing variable is often the future, not the present	22
14	Anxiety and shame can keep people from the help they need	23
15	Financial AI is scaling fast, mostly for service and speed	24
16	The system often knows the file better than the consumer	25
17	A completed file is not a reconstructable decision	26
18	A design philosophy, not a claim of therapy	28
19	Private digital settings can lower some disclosure barriers	29
20	Social and technical privacy pull in opposite directions	30
21	Concluding too early makes a chatbot feel judgmental	31
22	Conversation keeps people engaged with complex material	32
23	Front-loading context changes the professional meeting	33
24	Relationship and memory affect whether people come back	34
25	Coaching and decision aids support the human side	35
26	Conversation can make a wrong answer feel more trustworthy	36
27	Safety architecture matters more than a friendly tone	37
28	Future-Level Hazard Assessment: a financial prevention model	39
29	Compare what consumers say with what their finances show	40
30	Future hazards should be personal and concrete	41
31	Controls turn the assessment from fear into design	42
32	Residual risk and change triggers keep the model honest	43
33	A mortgage can qualify and still break the family's own limits	44
34	Give the professional a brief, not a raw transcript	45
35	Privacy is part of the intervention	47
36	Consumer-driven banking can make the audit more factual	48
37	A better handoff raises the first conversation	49
38	Evidence can support suitability without deciding it	50
39	One flow joins the consumer, the professional and the evidence	51
40	The first pilot tests three hypotheses, not the whole vision	53
41	Measure decision quality, not just engagement	54
42	Kill tests show what would disprove the model	55

EXECUTIVE SUMMARY

Prevent the harm before the decision is made.

The central proposition of this paper is simple: high-risk industries improved safety by moving risk discovery upstream. Instead of waiting for an incident to reveal a hazard, modern safety systems identify foreseeable hazards before work, estimate likelihood and consequence, establish controls, reassess residual risk and stop or adapt when conditions change. OSHA and CCOHS describe this proactive cycle directly. [1–5]

Major financial decisions have a different consequence profile, but they share a structural problem: the quality of the outcome depends on information that may be incomplete, uncomfortable to disclose, poorly understood, or discovered too late. Canadian regulators already require meaningful KYC, risk-capacity analysis and suitability reasoning, yet a 2025 CSA/CIRO review of 105 firms found recurring weaknesses in risk profiling, financial-circumstances collection, updates after life changes and documentation of why a recommendation was suitable. [10]

Counselling-centred AI is proposed as the interaction layer, not the decision-maker. The supporting research does not show that "AI gives better financial advice." It shows that private digital interaction can sometimes reduce evaluation fear, conversational systems can increase engagement, structured digital intake can front-load context for professionals, and financial coaching and decision aids can improve knowledge, confidence or behaviour. The same literature also shows serious boundaries: chatbots can feel judgmental, users can over-trust conversational interfaces, and purpose-built safeguards matter. [18–35]

> **The best time to manage risk is before the event, not after it.**
>
> Chad Johnston · Founder, 4orm

The thesis in one page.

EXHIBIT 1 · THE THESIS IN ONE PAGE

INDUSTRIAL PREVENTION	PROPOSED FINANCIAL ANALOGUE
Identify the work and hazards	Understand the person, the decision and foreseeable future stressors
Likelihood × severity	Probability × consequence to the person's financial life
Controls before work	Safeguards, alternatives, buffers, verification and professional review
Residual risk	What remains after controls and whether the person is still comfortable proceeding
Stop / reassess when conditions change	Revisit the decision when income, rates, family circumstances or goals change
Incident record and learning	Decision record showing what was understood, considered, changed and agreed

Source: 4orm analysis, drawing on OSHA and CCOHS [1–5].

Can prevention improve major financial decisions?

Can the preventive discipline that changed industrial safety improve the way people make major financial decisions? The analogy is deliberately constrained. A mortgage is not an industrial worksite. Financial loss is not equivalent to physical injury or death. Human lives, values and relationships cannot be reduced to a matrix.

The research question is narrower and more defensible: can the transferable mechanisms of prevention - early hazard identification, consequence thinking, controls, reassessment and evidence - be redesigned around the privacy, autonomy and complexity of a financial decision?

EXHIBIT 2 · WHAT MUST BE TRUE FOR THE THESIS TO HOLD

TEST	QUESTION THAT MUST BE ANSWERED
Decision quality	Does deeper discovery improve what the person understands and the choices they consider?
Counselling mechanism	Does reflective questioning improve comprehension or merely increase conversation length?
Disclosure	Does a private digital setting surface relevant context without encouraging unnecessary disclosure?
Professional value	Does the consumer-approved handoff help the professional understand the person and reduce rework?
Autonomy	Can the system support reflection without steering the person toward a commercial outcome?
Evidence	Can the record show how the recommendation related to the person without claiming the software itself determined suitability?
Outcomes	Do pilots reduce preventable misunderstanding, rework or unsuitable decisions enough to justify implementation?

Source: 4orm research design.

This paper therefore searches for both confirming and disconfirming evidence. The 4orm thesis should be narrowed or rejected where controlled testing shows no improvement in comprehension, professionally useful context or decision quality - or where privacy, pressure, over-reliance or implementation burden worsen. This challenge discipline carries forward from 4orm's earlier research on financial decision quality and evidence, which explicitly treated "what would weaken our thesis" as part of the research design.

FOUNDER PERSPECTIVE

The timing of risk discovery mattered most.

4orm's founder thesis grew from experience in two very different environments: critical life-safety work in oil and gas, and later work with people navigating trading, investing, fraud concerns and major financial decisions. The common lesson was not that the risks were equivalent. It was that the timing of risk discovery mattered.

> **The industries are different. The principle is the same: understand first, identify risk early, put the right information and controls in place, make the decision with greater clarity and preserve the evidence of what happened.**
>
> Chad Johnston · Founder, 4orm

In industrial work, the discipline is visible: before the task begins, the worker and team identify conditions that could cause harm, assess their significance and confirm that safeguards are in place. In financial decisions, many of the most important facts are equally time-sensitive but less visible: whether income is stable, what happens during parental leave, whether emergency savings will disappear, whether the person is relying on overtime, how they behave when cash flow tightens, or whether the recommended commitment leaves room for future obligations.

> **Help people discover and be open and transparent with themselves about the risks around them - the future of their finances and the decisions they are making now. Then let them choose what matters to bring to the professional serving them.**
>
> Chad Johnston · Founder reflection, edited for clarity

That is the foundation for Future-Level Hazard Assessment: a private, consumer-controlled pre-decision process that helps the person understand the present, explore plausible future conditions, identify controls and arrive at the professional conversation better prepared.

01 · PREVENTING HARM IN HIGH-RISK INDUSTRIES

High-risk industries learned to prevent harm.

56% reduction in corporate Total Recordable Injury Frequency at Canadian Natural since 2021

The move from reacting to incidents toward identifying hazards, controlling exposure and reassessing risk before work begins.

> **"The best time to manage risk is before the event, not after it."**
>
> Chad Johnston

KEY TAKEAWAYS

- Modern safety systems identify foreseeable hazards before work and reassess residual risk.
- Controls matter more than warnings: protection works best close to the source.
- The transferable lesson is prevention architecture, not consequence equivalence.

Source: Canadian Natural Resources, company-reported [8].

SECTION 01

Safety improved when risk discovery moved upstream.

OSHA describes traditional approaches to workplace hazards as often reactive: action is taken after a worker is injured or becomes sick, after a new standard appears, or after an inspection finds a problem. Its recommended model is the opposite - systematically find and fix hazards before they cause harm. [1]

That shift is broader than a single form. It represents a change in operating philosophy: from measuring safety mainly through events that already occurred to creating systems that continuously identify conditions that could cause harm. Near misses, changing work conditions, worker observations, task planning and inspection become inputs before the next incident.

EXHIBIT 3 · REACTIVE VERSUS PREVENTIVE SAFETY

REACTIVE SEQUENCE	PREVENTIVE SEQUENCE
Work begins	Understand the task
Incident or near miss occurs	Identify hazards and changing conditions
Investigation reconstructs causes	Assess likelihood and severity
Corrective action is added	Eliminate or reduce exposure through controls
Procedure is updated	Confirm residual risk before starting
Learning follows harm	Monitor and reassess as work changes

Source: OSHA [1]; 4orm analysis.

Boundary: modern safety still uses incident investigation and lagging indicators. The change is not "reactive versus proactive" as mutually exclusive systems; mature systems use both, with greater emphasis on prevention.

SECTION 02

Field-level assessment shows the risk before work begins.

CCOHS describes field-level risk assessment (FLRA) as a method commonly used where work conditions are dynamic and changing. Each step of the task is written down, hazards are identified, risk is assessed based on likelihood and severity, and the team decides whether existing controls are adequate or more measures are required before work begins. [2]

EXHIBIT 4 · THE PREVENTIVE LOOP

01 **Define the task** What are we about to do?	02 **Identify hazards** What could cause harm?	03 **Assess risk** How likely is it, and how severe could the consequence be?	04 **Select controls** What can eliminate, reduce or isolate the exposure?
05 **Reassess** What residual risk remains after controls?	06 **Decide** Are conditions acceptable to proceed?	07 **Monitor** What changed, and do we need to stop or reassess?	

Source: CCOHS [2].

The practical strength is not mathematical precision. It is structured attention. A worker who has repeated a task hundreds of times is required to look again at today's conditions, identify what is different and make risk visible to the team. That reduces the chance that familiarity becomes complacency.

> **The form was never the safety system. The point was to slow the work down long enough to see the hazard before the hazard controlled the outcome.**
>
> Chad Johnston · Founder reflection

SECTION 03

Likelihood × severity turns concern into a decision.

Risk assessment separates two questions that people often collapse: how likely is an event, and how serious would the consequence be if it occurred? CCOHS expresses the common logic as risk = probability × severity and notes that matrices are often used to prioritize action. [2]

EXHIBIT 5 · ILLUSTRATIVE RISK MATRIX

LIKELIHOOD	MINOR	MODERATE	MAJOR	SEVERE
Unlikely	Low	Low	Medium	Medium
Possible	Low	Medium	High	High
Likely	Medium	High	High	Critical
Very likely	Medium	High	Critical	Critical

Severity across the top; likelihood down the side.

Illustrative; CCOHS [2].

The matrix does not predict the future. It forces an explicit conversation about exposure. A low-probability event with catastrophic consequences may deserve stronger controls than a frequent inconvenience. Conversely, a high-likelihood event with manageable consequences may be accepted with monitoring.

The transferable principle for finance is important: “unlikely” does not mean “irrelevant,” and “affordable today” does not mean “resilient under plausible change.” A future-level assessment would use the same separation between probability and consequence, while avoiding false precision.

SECTION 04

Controls matter more than warnings.

Modern safety practice does not treat a warning as the strongest form of protection. The hierarchy of controls prioritizes eliminating or substituting hazards and using engineering controls before relying on administrative controls or personal protective equipment. CCOHS and NIOSH both emphasize controlling risk as close to the source as possible. [3–4]

EXHIBIT 6 · THE FINANCIAL ANALOGUE OF THE HIERARCHY OF CONTROLS

INDUSTRIAL CONTROL PRINCIPLE	POSSIBLE FINANCIAL ANALOGUE
Eliminate the hazard	Do not proceed with a decision whose downside cannot be carried or whose provider cannot be verified
Substitute	Choose a lower-risk product structure, smaller commitment or different timing
Engineering control	Automate buffers, payment limits, account protections or data verification
Administrative control	Document scenarios, disclosures, follow-up triggers and professional review
PPE / last-line protection	Emergency reserves, insurance or other protections that reduce impact after the risk materializes

Source: CCOHS [3]; NIOSH [4]; 4orm analysis.

The analogy is conceptual, not literal. The key insight is that education alone is weak if the conditions remain unsafe. Telling a consumer to “budget carefully” is not equivalent to structuring a payment that remains manageable under the scenarios the consumer says matter.

SECTION 05

Process safety widened the lens from the worker to the system.

OSHA's Process Safety Management standard was published in 1992 after catastrophic chemical-release incidents reinforced that complex hazards cannot be controlled through frontline behaviour alone. PSM requires an integrated management program that considers process hazards, controls, procedures, non-routine work, training, emergency response and management practices. [5]

OSHA defines process safety management as the proactive identification, evaluation and mitigation or prevention of hazardous releases. The deeper lesson is systems thinking: severe outcomes often emerge from interacting failures across design, information, procedures, equipment and management, not one isolated mistake.

EXHIBIT 7 · THE SYSTEMS LESSON

WEAK FRAMING	SYSTEMS FRAMING
"The worker made a mistake."	What conditions, controls, information and decisions made the mistake possible?
"The customer chose the product."	What did the customer understand, what alternatives were considered and what context shaped the choice?
"The disclosure was signed."	Was the material consequence understood, and what happened when information conflicted?
"The file is complete."	Can a reviewer reconstruct why the decision made sense at the time?

Source: 4orm analysis.

SECTION 06

Prevention through Design moves protection earlier.

NIOSH's Prevention through Design program argues that one of the most reliable ways to protect workers is to eliminate hazards and minimize risks early in design or redesign. If a hazard is designed out, the organization does not have to rely indefinitely on training, supervision or protective equipment to compensate for it. [4]

This is the furthest upstream form of prevention: do not wait for the person at the point of exposure to carry the entire burden. Design the environment so safer choices are easier and high-risk conditions are visible earlier.

> **The financial version of prevention through design is not another disclosure at the end. It is designing understanding, scenario testing and safeguards into the decision before the product is selected.**
>
> 4orm research synthesis

For 4orm, that means the pre-decision experience itself becomes a control. The system should surface missing information, contradictions and future pressure before the professional is asked to recommend a product, not only document what happened after the transaction closes.

SECTION 07

Frontline safety is continuous, not one-and-done.

Field-level assessments are commonly completed in addition to broader formal assessments. The same task may be reassessed when conditions, people, equipment or environment change. CCOHS explicitly recommends reviewing risk whenever conditions change and evaluating the residual risk that remains after controls. [2–3]

This distinction matters for finance because a point-in-time questionnaire can become stale. A mortgage that was reasonable at origination may become materially different after parental leave, job loss, retirement, divorce, renewal-rate change or new obligations. CIRO's 2025 review found firms sometimes knew of significant client changes but did not update KYC information or revisit suitability appropriately. [10]

EXHIBIT 8 · STATIC FORMS VERSUS LIVING RISK CONTEXT

STATIC RECORD	LIVING ASSESSMENT
Captured at one moment	Updated when relevant conditions change
Answers treated as fields	Answers connected to goals, dependencies and consequences
Completion proves the process occurred	Meaningful interaction and rationale are preserved
Change discovered during review	Change can trigger reassessment before the next decision

Source: CCOHS [2–3]; CSA / CIRO [10]; 4orm analysis.

SECTION 08

PCL shows prevention operating at scale.

PCL describes safety as embedded in how work is planned and executed rather than a separate activity. Its public safety materials emphasize frontline leadership, pre-job safety instruction, hazard inspection, detailed project safety plans and continuous improvement. [6]

46.9M
HOURS WORKED IN 2024 ACROSS PCL AND TRADE PARTNERS

0
LOST-TIME INCIDENTS REPORTED BY PCL FOR THAT YEAR

119 yrs
FIRST YEAR IN PCL HISTORY WITH THAT RESULT

PCL's public reporting is company-reported evidence and should not be used to attribute the outcome to any single control. It does, however, demonstrate the operating maturity possible when planning, formal procedures, audits, worker engagement and technology are combined. PCL also describes machine-learning hazard inspection and digital pre-job safety instruction tools designed to improve data quality and prevent incidents. [6]

The important lesson for finance is not the headline safety rate. It is the operating model: prevention is part of the workflow, workers participate in identifying risk, technology supports rather than replaces judgment, and the organization continuously learns from both leading and lagging signals.

SECTION 09

Syncrude and Canadian Natural make frontline control explicit.

Syncrude's current site-specific contractor package states that contract workers must follow their company Field Level Hazard Assessment process. Syncrude employees use a Critical Hazard Assessment Talk (CHAT) to identify and eliminate workplace practices and at-risk conditions that could lead to an incident, and the site maintains a hazard inventory reviewed annually. [7]

Canadian Natural describes a Safety Management System with 15 industry-recognized elements, continuous improvement and frontline engagement. It reports a **56%** reduction in corporate Total Recordable Injury Frequency since 2021 for employees and contractors. At Horizon, contractor Bouchier worked more than 1.75 million hours over 12 consecutive months without a recordable injury. [8]

EXHIBIT 9 · WHAT THESE OPERATING SYSTEMS HAVE IN COMMON

MECHANISM	OBSERVED IN INDUSTRIAL PRACTICE
Frontline participation	Workers and supervisors participate in hazard identification and control
Structured assessment	Tasks and changing conditions are assessed before or during work
Escalation rights	Workers can stop or refuse dangerous work
System-level inventory	Hazards are tracked beyond one task or one day
Continuous improvement	Performance data and incidents feed back into prevention

Source: Syncrude [7]; Canadian Natural [8]; PCL [6].

Boundary: company safety metrics are self-reported and reflect entire systems. This paper does not infer that FLHA alone caused reported injury reductions.

SECTION 10

The lesson is prevention, not equal consequences.

The industrial research supports five mechanisms worth carrying into finance: identify risk before exposure; distinguish likelihood from severity; prefer controls over warnings; reassess when conditions change; and preserve evidence of what was known and done. It does not support treating financial harm as equivalent to physical injury or death.

EXHIBIT 10 · TRANSFER WITH BOUNDARIES

CARRY FORWARD	DO NOT CARRY FORWARD
Pre-decision hazard identification	The assumption that every financial risk can be quantified precisely
Scenario and consequence thinking	A workplace-style command hierarchy over a consumer's personal choices
Control selection and residual-risk review	The idea that one "safe" answer exists for every person
Stop / reassess triggers	Automatic rejection based solely on a score
Evidence of reasoning and change	A claim that software can guarantee suitability or prevent loss

Source: 4orm analysis.

> **Safety taught me to respect the risk before the work started. Finance needs the same respect for what a decision can do to someone's future.**
>
> Chad Johnston · Founder reflection

02 · THE PRE-DECISION RISK PROBLEM

Finance has a pre-decision risk problem.

105 registered firms in the 2025 CSA/CIRO review, which found recurring KYC and suitability weaknesses

Suitability requirements exist, but important context, future-state risk and evidence can still arrive too late or remain difficult to reconstruct.

> **"A recommendation can make sense on paper and still fail the life around it."**
>
> Founder perspective

KEY TAKEAWAYS

- The missing variable is often the future, not the present.
- Financial anxiety and shame can make the people who need help most likely to avoid it.
- A completed file is not the same as a reconstructable decision.

Source: CSA / CIRO Staff Notice 31-368 [10].

SECTION 11

The financial decision starts before the application.

A person does not become a financial decision at the moment a form is opened. By then, they may already have anchored on a home, vehicle, investment or insurance outcome; discussed it with family; built expectations around a monthly payment; or decided what they are willing to sacrifice to make it happen.

Traditional workflows are strongest at capturing the transaction: identity, income, assets, product attributes, disclosures and approvals. They are weaker at preserving the private reasoning that shaped the decision before the application and the future conditions that could change whether the commitment remains comfortable.

EXHIBIT 11 · QUALIFICATION ANSWERS A NARROWER QUESTION THAN DECISION RESILIENCE

QUALIFICATION ASKS	DECISION RESILIENCE ALSO ASKS
Can this person meet today's underwriting criteria?	What future conditions could make the commitment uncomfortable or unsustainable?
What income/assets can be verified?	How stable are those inputs and what does the person refuse to sacrifice?
What product fits the formal criteria?	Which trade-offs fit the person's goals, buffers and future plans?
What disclosures are required?	What material consequences does the person actually understand?

Source: 4orm analysis.

> **Qualification can tell you whether the numbers fit. It cannot, by itself, tell you whether the decision fits the person.**
>
> 4orm research synthesis

SECTION 12

Regulation expects understanding. Execution is uneven.

The 2025 CSA/CIRO Client Focused Reforms review examined 105 registered firms. Regulators found recurring deficiencies in collecting risk tolerance and risk capacity, gathering financial circumstances, updating KYC after significant life changes, resolving inconsistencies and documenting the basis for suitability determinations. [10]

EXHIBIT 12 · WHAT THE REGULATOR FOUND

OBSERVED ISSUE	WHY IT MATTERS TO FLHA
Risk tolerance and risk capacity not adequately separated	Willingness to accept risk can differ from ability to withstand loss
Financial circumstances insufficiently detailed	The downside cannot be evaluated without liquidity, obligations and external assets
Known life changes not reflected in KYC	A decision record can become stale while the person's life changes
Superficial checklists / thin rationale	Completion does not explain how the factors were weighed
Alternatives not adequately considered or documented	Suitability needs a reasoned comparison, not only a chosen product

Source: CSA / CIRO Staff Notice 31-368 [10].

In British Columbia, the new Mortgage Services Act framework similarly requires mortgage licensees to take reasonable steps to determine suitability before presenting an option, considering the borrower's needs, circumstances and material risks. [11]

SECTION 13

The missing variable is often the future, not the present.

Financial suitability frameworks already consider time horizon, liquidity, risk capacity and personal circumstances. The opportunity is to make future conditions more explicit and more understandable to the person before commitment.

EXHIBIT 13 · EXAMPLES OF FUTURE-STATE HAZARDS

DECISION	FUTURE CONDITION WORTH TESTING
Mortgage	Renewal-rate increase; parental leave; childcare; income reduction; home repair; vehicle replacement
Auto finance	Insurance increase; maintenance; reduced commute reimbursement; job change; overlapping debt
Investment	Liquidity need arrives early; loss exceeds emotional or financial capacity; concentration grows; income falls
Insurance	Coverage assumptions no longer match family obligations; affordability changes; exclusions misunderstood
Small-business lending	Revenue seasonality worsens; key customer is lost; owner draw increases; refinancing conditions tighten

Source: 4orm analysis.

A future scenario is not a forecast. Its purpose is to reveal dependencies. If a mortgage only works when overtime remains constant, rates do not rise and no emergency reserve is used, that dependency should be visible before the person commits.

SECTION 14

Anxiety and shame can keep people from the help they need.

Financial decisions are not neutral data-entry exercises. A 2021 research program across six studies and **9,110** participants found that financial shame can drive withdrawal and disengagement, which in turn can worsen financial hardship. [15]

A separate two-study paper found participants with high financial anxiety were 2.75 times more likely to choose financial avoidance than those with low financial anxiety. [16] Financial adviser anxiety research also identifies embarrassment, worry and apprehension as barriers to seeking professional advice. [17]

9,110
PARTICIPANTS ACROSS FINANCIAL-SHAME STUDIES

2.75×
HIGHER LIKELIHOOD OF FINANCIAL AVOIDANCE IN HIGH-ANXIETY GROUP

1 problem
THE PROFESSIONAL MEETING ITSELF CAN BECOME A BARRIER

This creates a design problem: more disclosure requirements do not necessarily produce better disclosure. The person may need a lower-pressure place to understand their own situation before they can communicate it effectively to someone else.

SECTION 15

Financial AI is scaling fast, mostly for service and speed.

Banks and financial platforms are already using conversational systems at massive scale. Bank of America reported more than 3.2 billion Erica interactions since launch by March 2026. NatWest is trialling conversational spending insights that let customers explore transaction data through natural language. Intuit is expanding connected-account insights and agentic financial assistance. [36–38]

EXHIBIT 14 · WHERE CURRENT FINANCIAL AI IS STRONGEST

CAPABILITY	EXAMPLES	WHAT REMAINS DIFFERENT IN THE 4ORM THESIS
Service and navigation	Erica, Cora	Resolve questions and tasks efficiently
Spending insight	NatWest Spending Insights, Credit Karma	Explain patterns in transaction data
Product / action support	Bank and fintech agents	Help the user complete a financial task
Proposed 4orm layer	Counselling-centred FLHA	Explore goals, future hazards, controls and decision readiness before recommendation

Source: Bank of America [36]; NatWest Group [37]; Intuit [38]; 4orm analysis.

The gap is not that finance lacks AI. The gap this paper investigates is whether AI can be deliberately designed to help the consumer understand themselves and the future decision before the professional recommendation begins - while preserving a structured, permissioned handoff.

SECTION 16

The system often knows the file better than the consumer.

A well-run financial workflow can verify identity, income and product terms and still miss the story that makes those facts meaningful. A person may qualify based on current income while planning parental leave. They may have sufficient assets but refuse to spend an emergency reserve. They may tolerate investment volatility in theory but be unable to absorb a loss before a known cash need.

CIRO's own guidance distinguishes risk tolerance from risk capacity and says financial circumstances should include income, liquidity needs, assets, net worth and leverage. It also requires meaningful interaction and documentation of how inconsistencies are resolved. [10]

> **The person understands the financial decision better, and the business understands the person better. That is the relationship we are trying to improve.**
>
> Chad Johnston · Founder, 4orm

Future-Level Hazard Assessment is intended to add context to these obligations, not replace them. The output should make the professional's job easier: a clearer view of the person's goals, dependencies, scenario boundaries and unresolved questions before product selection.

SECTION 17

A completed file is not a reconstructable decision.

4orm's earlier Canadian research found a recurring evidence problem: regulatory expectations require firms to understand the client and support suitability, while supervision and operations can still rely heavily on documents, checklists and later reconstruction. The 2025 CSA/CIRO review reinforces that distinction: some firms had little or no documentation showing the basis for suitability, while others relied on superficial tools without evidence of how factors were considered. [10]

EXHIBIT 15 · THE EVIDENCE GAP

WHAT A FILE MAY SHOW	WHAT A FUTURE REVIEWER MAY STILL NEED
KYC values	How the values were discussed and whether they conflicted
Signed disclosure	What material consequence the person understood
Selected product	What alternatives were considered and why rejected
Approval timestamp	What changed between discovery and commitment
Professional note	Which source facts support the reasoning and what remained uncertain

Source: CSA / CIRO [10]; 4orm analysis.

The proposed evidence layer should therefore capture the sequence, not only the endpoint: what the person said mattered, which hazards were explored, what controls changed the risk, what the professional explained, and what the person ultimately chose to share and approve.

03 · LESSONS FROM COUNSELLING-CENTRED SYSTEMS

What counselling-centred systems can teach finance.

73.4% accepted an objectively incorrect recommendation from a conversational robo-adviser, versus 40.0% without

Evidence from disclosure psychology, motivational interviewing, digital intake, decision aids and purpose-built conversational systems.

> "The goal is not to give the person an answer faster. It is to help them understand what could change the answer."
>
> Chad Johnston

KEY TAKEAWAYS

- Private digital interaction can lower some disclosure barriers, if technical trust is strong enough.
- A chatbot can feel more judgmental when it concludes before it understands.
- Conversation can make a wrong answer feel more trustworthy, so understanding and persuasion must be separated.

Source: Hilbrand et al. [31].

SECTION 18

A design philosophy, not a claim of therapy.

“Counselling-Centred AI” is a proposed 4orm term. It does not mean the system is a therapist or that financial decisions should be medicalized. It describes a non-directive conversational approach that borrows transferable interaction principles: reflective questioning, values clarification, context seeking, scenario exploration, summarization, checking understanding and preserving user agency.

EXHIBIT 16 · WHAT THE SYSTEM SHOULD DO - AND NOT DO

COUNSELLING-CENTRED BEHAVIOUR	BOUNDARY
Ask before concluding	Do not infer suitability from thin data
Reflect what the person said matters	Do not flatter or simply agree
Explore ambivalence and trade-offs	Do not steer toward a product or transaction
Seek missing context	Do not treat silence as evidence
Surface contradictions respectfully	Do not decide which value should win
Summarize and ask for correction	Do not convert private exploration into business data without permission

Source: 4orm analysis.

Motivational interviewing is especially relevant because it emphasizes autonomy, reflective listening and eliciting a person’s own reasons rather than directing them. A 2026 pragmatic randomized trial found a motivational-interviewing chatbot could improve lifestyle outcomes in primary care, supporting the broader idea that structured, autonomy-preserving conversation can influence behaviour. [26]

SECTION 19

Private digital settings can lower some disclosure barriers.

The strongest foundation predates modern generative AI. A meta-analysis of 460 effects across **125,672** participants found computerized self-administered surveys produced more reporting of socially undesirable behaviours than comparable paper surveys, especially for highly sensitive topics. [18]

Lucas and colleagues experimentally manipulated whether participants believed a virtual interviewer was controlled by a computer or a human. Believing it was computer-controlled reduced evaluation fears and impression management and increased objectively rated disclosure. [19]

A 2026 five-experiment study with **1,461** participants found higher intended disclosure to an AI psychotherapist than to human alternatives, with reduced fear of negative evaluation mediating the effect. The study also found an important boundary: making privacy-security risk salient weakened the AI disclosure advantage. [20]

> **The opportunity is not "people tell AI everything." It is that lower social pressure may give some people more room to think honestly - if technical trust is strong enough.**
>
> 4orm research synthesis

SECTION 20

Social and technical privacy pull in opposite directions.

A private digital setting can reduce fear of being judged by another person while simultaneously increasing concern about data collection, retention and misuse. Finance-specific evidence demonstrates the tension. In three experiments with **3,346** participants, placing financial support tools inside personal customer portals increased perceived privacy risk but also increased positive affect enough to increase willingness to disclose. [21]

EXHIBIT 17 · THE DISCLOSURE EQUATION IS CONDITIONAL

CONDITION	LIKELY EFFECT	DESIGN RESPONSE
Low social pressure + strong technical trust	More room for candid exploration	Private-first experience; clear purpose; visible controls
Low social pressure + weak technical trust	Withdrawal or guarded answers	Explain what is collected, retained and shared at the moment it matters
High warmth + commercial conflict	Trust can become persuasion	Separate exploration from sales optimization
Fast conclusions from thin context	System feels judgmental	Show what is known, missing and assumed before suggesting next steps

Source: Lucas et al. [19]; Technology in Society [20]; Demmers, Dellaert and Rotaru [21]; 4orm analysis.

PIPEDA's principles reinforce the product implication: identify purposes, obtain meaningful consent, limit collection, maintain accuracy, apply safeguards and support access and correction. [14]

SECTION 21

Concluding too early makes a chatbot feel judgmental.

The disclosure story is not one-directional. A 2025 MIS Quarterly paper conducted four experiments and a qualitative study and found participants perceived a text-based chatbot as more judgmental than a human mental-health professional even when the interactions were identical. The perception reduced willingness to use the service, disclose information and follow recommendations. [41]

Participants described “judgment” as a rush to conclusion without deeply grasping the issue or conveying that they had been heard. That mechanism maps directly to financial discovery. A system that labels a consumer “high risk” after a few answers may create resistance even if the label is statistically reasonable.

EXHIBIT 18 · DESIGN IMPLICATION: UNDERSTANDING BEFORE INTERPRETATION

POOR SEQUENCE	COUNSELLING-CENTRED SEQUENCE
Collect answer	Collect answer
Score immediately	Ask follow-up where context is missing
State conclusion	Reflect the person’s meaning back
Move to recommendation	Ask for correction / confirmation
Explain after	Only then surface trade-offs, hazards or next questions

Source: Raimi et al. [41]; 4orm analysis.

> **If the system reaches a conclusion before the person feels understood, we have recreated the very barrier we were trying to remove.**
>
> 4orm research synthesis

SECTION 22

Conversation keeps people engaged with complex material.

In a six-week randomized trial with 540 participants, a generative-AI CBT experience produced 2.4 times more engagement sessions and 3.8 times longer engagement than static digital workbooks delivering comparable content. Clinical outcomes were comparable between groups; the correct conclusion is greater engagement, not 3.8 times greater treatment efficacy. [24]

2.4×	3.8×	N=540
MORE ENGAGEMENT SESSIONS	LONGER ENGAGEMENT DURATION	RANDOMIZED TRIAL PARTICIPANTS

The financial analogue is not therapy. It is sustained thinking. Many financial decisions require a person to work through costs, trade-offs, scenarios and unfamiliar terminology. A responsive conversation may keep the person engaged long enough to ask the second and third question that a static disclosure does not prompt.

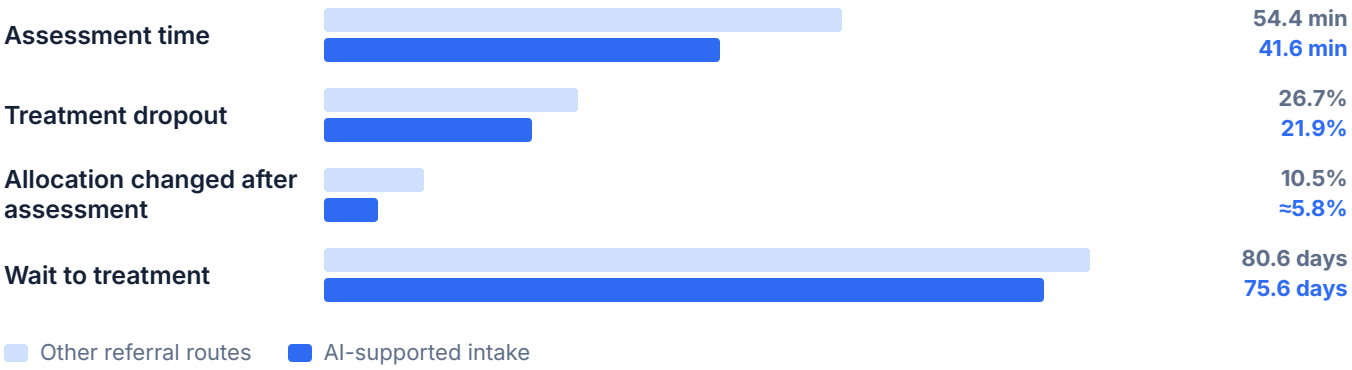
FCAC's 2026 randomized study offers a finance-specific signal: a short, behaviourally informed digital debt intervention helped participants reduce credit-card debt by **35%** more than the control group, illustrating that even brief, well-timed digital interventions can change financial behaviour in defined contexts. [39]

SECTION 23

Front-loading context changes the professional meeting.

The closest workflow analogue to 4orm is AI-supported intake before a human professional assessment. In an observational study of **64,862** NHS patients, AI-supported conversational intake was associated with 12.7 fewer assessment minutes, lower dropout, fewer treatment-allocation changes after assessment and a shorter wait to treatment. The researchers identified clinically relevant information supplied before assessment as an important mechanism. [22]

EXHIBIT 19 · AI-SUPPORTED INTAKE IN 64,862 PATIENTS



Reported differences: 12.7 min lower; 18% relative reduction in dropout; ≈45% relative reduction in changed allocation; 5 days lower wait.

Source: Rollwage et al. [22]. Figures as reported; healthcare is an adjacent workflow analogue.

Evidence boundary: healthcare is an adjacent workflow analogue, not proof that financial discovery will save the same time or improve suitability by the same amount.

The transferable hypothesis is operational: gather context before the professional meeting, structure it, let the person correct it, and pass only approved information so the professional begins at a higher level.

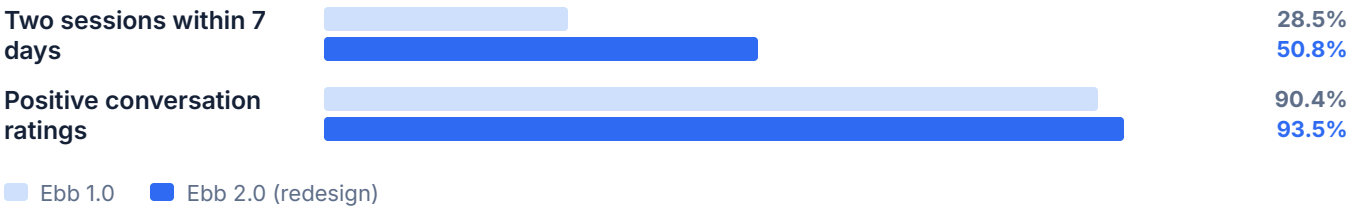
SECTION 24

Relationship and memory affect whether people come back.

Wysa research with **1,205** users found measurable therapeutic-alliance scores within days of use, with transcript evidence of gratitude, perceived impact and personification. Because company employees were among the authors and the setting was mental health, the result should be treated as a design signal, not proof of a financial relationship effect. [28]

Headspace's 2026 real-world study across **393,969** members compared two versions of its conversational AI companion. The redesign added stronger prompts, comprehensive memory, content recommendations and more robust safety detection. Seven-day two-session retention rose from **28.5%** to **50.8%**, and positive conversation ratings increased from **90.4%** to **93.5%**. [29]

EXHIBIT 20 · HEADSPACE EBB: TWO SESSIONS WITHIN SEVEN DAYS ROSE AFTER THE REDESIGN



Source: Callahan et al., JMIR Formative Research [29]; 393,969 members.

Source: Headspace / JMIR Formative Research. Observational product-version comparison; not a randomized causal estimate.

For 4orm, the important lesson is not anthropomorphism. It is continuity: a system that remembers what the consumer said mattered can ask better follow-up questions and detect when today's proposed decision conflicts with an earlier stated boundary.

SECTION 25

Coaching and decision aids support the human side.

The CFPB's randomized evaluation of financial coaching found measurable gains in money management, objective financial-health metrics such as savings, debt and credit scores, and subjective financial confidence and well-being. Coaching was personalized around each person's goals. [32]

A review of 34 systematic reviews on patient decision aids found outcomes across knowledge, participation, decisional conflict, regret, satisfaction and adherence. Healthcare is not finance, but the mechanism is directly relevant: structured information plus values clarification can help a person participate more meaningfully in a professional decision. [33]

EXHIBIT 21 · WHAT TRANSFERS INTO A FINANCIAL DECISION-READINESS MODEL

EVIDENCE FIELD	TRANSFERABLE MECHANISM
Financial coaching	Personalized goals, accountability, confidence and behaviour change
Decision aids	Make options, risks and trade-offs comparable
Motivational interviewing	Elicit the person's own reasons and preserve autonomy
Conversational intake	Gather structured context before professional assessment
Longitudinal memory	Track what changed instead of treating each conversation as isolated

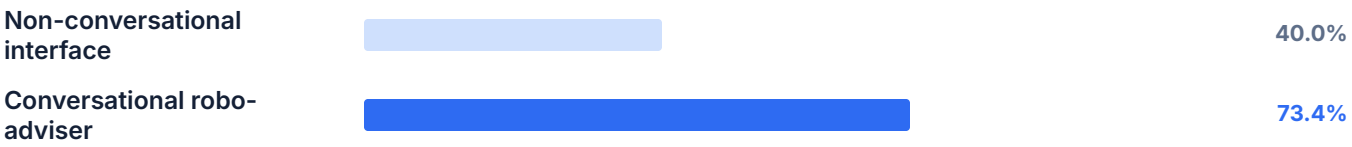
Source: CFPB [32]; decision aid review [33]; npj Digital Medicine [26]; 4orm analysis.

SECTION 26

Conversation can make a wrong answer feel more trustworthy.

A finance-specific experiment found participants interacting with a conversational robo-adviser were significantly more likely to accept an objectively incorrect portfolio recommendation than participants using a non-conversational interface: **73.4%** versus **40.0%**. The effect appeared for both risk-averse and risk-seeking participants. [31]

EXHIBIT 22 · CONVERSATIONAL TRUST INCREASED ACCEPTANCE OF A WRONG RECOMMENDATION



Source: Hilbrand et al. [31]. The effect appeared for both risk-averse and risk-seeking participants.

This finding creates a non-negotiable boundary for counselling-centred financial systems: understanding and persuasion must be separated. The interface should not exploit warmth, personalization or memory to push a product, increase conversion or make the system's recommendation feel socially costly to reject.

EXHIBIT 23 · ROLE SEPARATION

01	02	03
4orm / counselling layer	**Regulated professional**	**Consumer**
Explore, clarify, simulate, surface uncertainty	Recommend, advise, determine suitability, execute where permitted	Choose, correct, grant permission, decide

Source: 4orm analysis.

SECTION 27

Safety architecture matters more than a friendly tone.

The 2026 scoping review of generative-AI mental-health chatbots found convenience, personalization and perceived empathy can support user experience, while inaccurate or contextually misaligned outputs can erode trust. A separate 2026 scoping review identified 119 papers on potential mental-health harms including hallucination, sycophancy, dependence, privacy risks and high-acuity failure modes. [34–35]

Headspace's public Ebb architecture illustrates the product discipline: **100%** of messages are passed through risk classification, higher-acuity messages can enter clinician review, guardrails limit out-of-scope behaviour, and the company uses pre-release evaluation, red teaming and post-release monitoring. [40]

EXHIBIT 24 · FINANCIAL ANALOGUE OF SAFETY-BY-DESIGN

MENTAL-HEALTH SAFEGUARD	FINANCIAL-DECISION ANALOGUE
Risk classifier	Detect fraud signals, coercion, severe distress, missing professional jurisdiction or unsafe advice requests
Scope boundary	Do not provide regulated advice the system is not authorized to give
Human escalation	Route material suitability, fraud or conflict concerns to a qualified professional
Pre-release evaluation	Test scenarios, edge cases, bias, sycophancy and privacy failure before deployment
Post-release monitoring	Track harmful patterns, overrides, complaints and misunderstood outputs

Source: Headspace [40]; 4orm analysis.

04 · THE FUTURE-LEVEL HAZARD ASSESSMENT

The Future-Level Hazard Assessment.

35% more credit-card debt reduction than the control group after a short digital intervention

A proposed 4orm model for identifying foreseeable financial hazards, applying controls and reassessing the decision before commitment.

> **"The industries are different. The principle is the same: understand first, identify risk early, put the right controls in place."**
>
> Chad Johnston

KEY TAKEAWAYS

- FLHA is a repeatable pre-decision process for finding what could make a commitment stop working.
- Controls turn the assessment from fear into design; residual risk and change triggers keep it honest.
- The professional should receive a decision brief, not a raw transcript.

Source: Financial Consumer Agency of Canada, 2026 [39].

SECTION 28

Future-Level Hazard Assessment: a financial prevention model.

Future-Level Hazard Assessment (FLHA) is a proposed 4orm framework inspired by field-level hazard assessment. The purpose is not to label a person or predict their life. It is to create a repeatable pre-decision process for identifying the conditions that could make a financial commitment stop working, then exploring controls before commitment.

EXHIBIT 25 · THE FLHA SEQUENCE

01 **Decision** What financial decision am I considering and what am I trying to accomplish?	02 **Current state** What does my actual financial behaviour show today?	03 **Future hazards** What foreseeable changes could make this decision harder?	04 **Risk** How likely is each scenario, and how significant would the consequence be for me?
05 **Controls** What could eliminate, reduce, transfer or absorb the risk?	06 **Residual risk** What remains after the controls?	07 **Readiness** Am I still comfortable proceeding, and what does my professional need to understand?	08 **Reassessment** What future change should trigger a new review?

Source: 4orm analysis.

Source: FLHA is a 4orm concept. It is not an established Canadian financial-regulatory term and is not intended to replace KYC, suitability, underwriting or professional judgment.

SECTION 29

Compare what consumers say with what their finances show.

A consumer may describe themselves as a strong saver while account data shows repeated month-end shortfalls; or they may appear cash-constrained because they deliberately move large amounts into savings. The point of an audit is not to “catch” the consumer. It is to give them evidence about their own patterns and a chance to explain context.

EXHIBIT 26 · POTENTIAL CONSUMER-PERMISSIONED AUDIT SIGNALS

SIGNAL	QUESTION IT CAN SUPPORT
Payment consistency	Are recurring obligations usually paid comfortably and on time?
Cash-flow variability	How much does monthly income or spending move?
Savings pattern	Is saving regular, irregular or depleted after major expenses?
Emergency reserve	How many months of essential obligations can current liquid savings cover?
Debt service	How much of current cash flow is already committed?
Discretionary flexibility	Which expenses are truly flexible if conditions worsen?
Income concentration	How dependent is the household on overtime, one payer or one source?

Source: 4orm analysis.

Canada’s emerging consumer-driven banking framework is important because it is explicitly designed to let consumers securely share account, balance, transaction and product data with approved providers of their choice. The policy objective includes giving people greater control and a more complete financial picture. [13]

SECTION 30

Future hazards should be personal and concrete.

The useful financial hazard is not “market risk” in the abstract. It is a condition the person can recognize and discuss: “If rates renew three points higher while we are on one income, this payment would consume the reserve we said we wanted to protect.”

EXHIBIT 27 · A PRACTICAL HAZARD TAXONOMY

HAZARD FAMILY	EXAMPLES
Income	Job loss, reduced overtime, commission variability, business slowdown, parental leave
Cost of living	Childcare, rent, food, utilities, insurance, elder support
Debt / rates	Renewal shock, variable-rate increase, revolving debt growth, refinancing constraints
Assets / liquidity	Emergency repair, vehicle replacement, illiquid investment, market drawdown
Life transition	Marriage, separation, retirement, disability, relocation, education
Behaviour	Impulse spending, recurring overdraft, inconsistent saving, avoidance of statements
Product / counterparty	Fees, lock-ins, renewal terms, exclusions, fraud or unverifiable provider claims

Source: 4orm analysis.

The system should also ask which risks the consumer does not want to optimize away. A person may knowingly accept a higher payment to live near family, or choose lower liquidity for a long-term goal. FLHA should make the trade-off explicit rather than impose a universal definition of “best.”

SECTION 31

Controls turn the assessment from fear into design.

The purpose of identifying hazards is not to create anxiety. In safety practice, the hazard assessment becomes useful when it leads to controls. The financial equivalent is to ask what could change the exposure and then reassess the residual risk.

EXHIBIT 28 · ILLUSTRATIVE MORTGAGE CONTROLS

HAZARD	INITIAL RISK	POSSIBLE CONTROL	RESIDUAL RISK
Renewal +3 percentage points	High	Smaller mortgage; fixed term; larger reserve; stress-test payment	Medium
One income during parental leave	High	Lower monthly commitment; dedicated leave reserve	Medium
Vehicle replacement within 24 months	Medium	Sinking fund; postpone purchase; reduce home budget	Low
Overtime falls	High	Base affordability on regular income	Medium
Emergency repair	Medium	Preserve six-month liquid reserve	Low

Illustrative; 4orm analysis.

A control can change likelihood, consequence or both. The consumer should see the “before” and “after” side by side. If residual risk remains high, the response is not automatically “decline.” It may be pause, seek professional review, change the structure, accept knowingly, or choose not to proceed.

SECTION 32

Residual risk and change triggers keep the model honest.

Safety assessment does not end when a control is written down. CCOHS recommends reassessing risk after controls and when conditions change. [2–3] Future-Level Hazard Assessment should do the same.

EXHIBIT 29 · TWO OUTPUTS THAT MATTER

RESIDUAL-RISK STATEMENT	CHANGE TRIGGER
“This mortgage remains comfortable if renewal rates rise two points, but becomes tight above three.”	Reassess 6–9 months before renewal or when income falls >**10%**
“The plan preserves six months of savings unless the vehicle must be replaced.”	Reassess if the vehicle fund falls below target
“The investment fits the time horizon only if the down-payment date remains flexible.”	Reassess when a purchase date becomes fixed

Illustrative; 4orm analysis.

This creates a longitudinal record of decision conditions. The system is no longer asking only, “Was the product suitable on the day it was recommended?” It can also preserve, “What conditions made it suitable, and which future changes were expected to trigger a review?”

SECTION 33

A mortgage can qualify and still break the family's own limits.

Illustrative household: two earners, one planning parental leave within 18 months, variable overtime, **$42,000** in liquid savings, a vehicle expected to need replacement within two years, and a stated goal to preserve at least six months of essential expenses.

EXHIBIT 30 · FUTURE-LEVEL HAZARD ASSESSMENT EXAMPLE

QUESTION	CURRENT ANSWER	WHAT CHANGES AFTER EXPLORATION
What must the decision achieve?	Buy the desired home	Also preserve family flexibility and emergency reserve
What does behaviour show?	Payments current	Savings strong but monthly surplus depends partly on overtime
Key future hazard?	"Rates might rise"	Rate renewal + parental leave + vehicle replacement interact
Control?	Use more savings as down payment	Consumer rejects because it violates reserve goal
Alternative?	Smaller mortgage / lower price / larger leave reserve	Reduces residual risk without sacrificing the stated buffer
Professional handoff	Income + assets	Adds the person's explicit boundaries, scenarios and unresolved questions

Illustrative household; 4orm analysis.

> **The point is not to tell someone they cannot buy the house. It is to let them see the future they are signing up for before they sign up for it.**
>
> Chad Johnston · Founder reflection

SECTION 34

Give the professional a brief, not a raw transcript.

The output of private exploration should be structured, reviewable and controlled by the consumer. Long transcripts create burden and privacy risk. A decision brief should carry the meaning necessary for the professional to serve the person better.

EXHIBIT 31 · CONSUMER-APPROVED DECISION BRIEF

SECTION	EXAMPLE
What I want	Buy a home while preserving family flexibility
What matters	Do not rely on overtime; keep six months of essential savings untouched
Risks I explored	Parental leave, renewal shock, childcare, vehicle replacement
Controls I am comfortable with	Lower monthly payment; dedicated leave reserve; stress-test at higher rate
What I understand	A lower initial payment can still create higher future risk
Questions I still have	Fixed vs variable; portability; prepayment flexibility
What I chose to share	Selected context, scenario results and account-derived summaries

Illustrative; 4orm analysis.

The design goal is a better-prepared person, not merely more data. The professional should be able to see where the information came from, what the person confirmed, what remains private and what still requires judgment.

05 · PRIVATE FIRST, SHARED BY CHOICE

Private first.
Shared by choice.

12.7 min fewer assessment minutes with AI-supported intake across 64,862 NHS patients

Consumer-controlled exploration, permissioned data and a professional handoff that improves understanding without turning private reflection into surveillance.

> **"The person should understand the decision better, and the business should understand the person better."**
>
> Chad Johnston

KEY TAKEAWAYS

- Privacy is part of the intervention, not a legal appendix.
- The consumer chooses which findings move into the professional workflow.
- The evidence layer can support suitability without claiming to determine it.

Source: Rollwage et al., observational study [22].

SECTION 35

Privacy is part of the intervention.

If candid exploration depends partly on lower fear of judgment, the product cannot immediately convert every reflection into data for the company selling the product. Privacy, autonomy and technical trust are therefore functional requirements of the model, not merely compliance obligations.

EXHIBIT 32 · PRIVATE FIRST, PERMISSIONED SHARING

CONSUMER CONTROL	PRODUCT REQUIREMENT
Purpose	Explain why each category of information is being requested
Choice	Let the person skip optional questions and still continue where safe
Review	Show what the system thinks it learned and allow correction
Selective sharing	Let the person choose which findings move into the professional workflow
Withdrawal	Allow future optional permissions to be withdrawn, subject to lawful retention duties
Access / export	Let the person retrieve the structured record they contributed

Source: PIPEDA principles [14]; 4orm analysis.

These controls align with PIPEDA principles of identified purpose, consent, limiting collection, accuracy, safeguards and individual access. [14] They also reduce a strategic risk: a system cannot credibly be “counselling-centred” if private vulnerability is quietly repurposed for sales optimization.

SECTION 36

Consumer-driven banking can make the audit more factual.

Budget 2025's consumer-driven banking framework is designed to let Canadians direct approved providers to receive financial data through secure APIs. In-scope data includes account information, balances, transactions and product data for deposit, payment, investment and lending accounts. [13]

The framework's stated goals include greater consumer control, a more complete financial picture and improved financial outcomes. That creates a policy path for an FLHA system to analyze real financial patterns without asking a consumer to upload screenshots or share online-banking credentials.

EXHIBIT 33 · DATA SHOULD ANSWER A CONSUMER QUESTION

DATA SIGNAL	CONSUMER-FACING QUESTION
Transaction history	What does my real monthly spending look like, not my estimate?
Balances	How much buffer do I actually keep through the month?
Debt payments	How much cash flow is already committed?
Savings flows	Do I save consistently or only in low-expense months?
Income deposits	How variable is my income, and how much depends on irregular sources?

Source: Department of Finance Canada [13]; 4orm analysis.

Boundary: Canada's consumer-driven banking framework is still being implemented. Product availability, accreditation, permitted use, data scope and provincial interaction require current legal and technical review before deployment.

SECTION 37

A better handoff raises the first conversation.

The Limbic evidence suggests the value of front-loaded intake is not automation for its own sake. It is that the professional receives relevant context before assessment and can spend less time reconstructing basic information. [22]

EXHIBIT 34 · BEFORE AND AFTER THE HANDOFF

WITHOUT PRE-DECISION CONTEXT	WITH CONSUMER-APPROVED FLHA CONTEXT
First meeting discovers goals and basic constraints	First meeting begins with confirmed goals, hazards and boundaries
Professional asks generic scenario questions	Professional can focus on the highest-impact scenarios already surfaced
Information may be scattered across notes and forms	Key facts are structured with source and permission status
Rationale is documented after the recommendation	Reasoning can be connected to the context as it develops
Later reviewer reconstructs the story	Later reviewer can trace changes, explanations and approvals

Source: 4orm analysis.

The professional still owns professional judgment. The system's role is to improve preparation and preserve the consumer-approved context that informs that judgment.

SECTION 38

Evidence can support suitability without deciding it.

CIRO requires suitability determinations to consider KYC, product understanding, account impact, costs and a reasonable range of alternatives, and to put the client's interest first. Staff guidance says firms need documentation sufficient to demonstrate a reasonable basis for the determination. [10]

EXHIBIT 35 · HOW AN FLHA RECORD COULD SUPPORT THE PROFESSIONAL

SUITABILITY INPUT	POSSIBLE EVIDENCE CONTRIBUTION
Personal / financial circumstances	Consumer-confirmed context plus permissioned account-derived summaries
Risk capacity	Scenario-specific ability to absorb downside, not only a self-selected label
Objectives / time horizon	Goals and future events stated in the person's own language
Alternatives	Record of options, controls and why trade-offs were accepted or rejected
Rationale	Dated link between the professional's recommendation and the information available at the time
Change	Record of material new information and whether reassessment occurred

Source: CIRO [10]; 4orm analysis.

4orm should avoid the phrase "proves suitability" when it implies that software itself has made the regulated determination. A more defensible claim is that the system can help the business evidence the basis for its suitability reasoning, subject to professional review and the applicable regulatory regime.

SECTION 39

One flow joins the consumer, the professional and the evidence.

EXHIBIT 36 · PROPOSED 4ORM OPERATING FLOW

LAYER	ROLE
1 · Private exploration	Goals, concerns, trade-offs, questions, future scenarios
2 · Financial audit	Consumer-permissioned summaries of cash flow, payments, saving and obligations
3 · FLHA reasoning	Hazards, likelihood, consequence, controls and residual risk
4 · Consumer approval	Review, correction and selective sharing
5 · Professional workflow	Context enters CRM / LOS / adviser tools; professional completes discovery and recommendation
6 · Evidence layer	Source, version, permissions, explanations, alternatives, changes and confirmations linked together
7 · Reassessment	Material life or product changes trigger a new review rather than silently aging the record

Source: 4orm proposed architecture.

The long-term opportunity is not a larger data warehouse. It is a decision graph: a connected record of the facts, human meaning, professional reasoning and changes that explain why a decision made sense at a particular point in time.

> **Understanding before the decision. Evidence after it.**
>
> 4orm Founder North Star

06 · FROM HYPOTHESIS TO EVIDENCE

From hypothesis to evidence.

119 papers on potential mental-health harms of LLM chatbots in a 2026 scoping review

Pilot design, measures and kill tests that can prove, narrow or reject the Future-Level Hazard Assessment thesis.

> **“The next step is not a bigger claim. It is better evidence.”**
>
> 4orm research principle

KEY TAKEAWAYS

- The first pilot should test consumer understanding, professional context and decision evidence.
- Measure decision quality, not just engagement.
- Kill tests set release blockers that a strong satisfaction score cannot average away.

Source: Diel et al., npj Digital Medicine [35].

SECTION 40

The first pilot tests three hypotheses, not the whole vision.

EXHIBIT 37 · THREE TESTABLE HYPOTHESES

HYPOTHESIS	PREDICTION	PRIMARY TEST
H1 · Better consumer understanding	FLHA users can explain material risks, trade-offs and change triggers more accurately than baseline discovery users	Comprehension task scored against predefined concepts
H2 · Better professional context	Professionals receive more relevant, confirmed context before recommendation and spend less time reconstructing it	Time study + blinded assessment of context completeness
H3 · Better decision evidence	The final file more clearly links customer circumstances, alternatives, explanations and recommendation rationale	Independent file-review rubric

Source: 4orm research design.

Mortgage brokering is a practical starting point because the decision is consequential, future affordability matters, Canadian suitability requirements are explicit, and the workflow already contains income, asset and product data that can be compared with consumer-defined goals and scenarios.

A staged design should begin with historical or simulated cases, move to supervised live use, and pre-register the primary measures before broad deployment. Adoption and satisfaction are secondary; the main question is whether the process improves understanding, context and evidence without creating new harm.

SECTION 41

Measure decision quality, not just engagement.

EXHIBIT 38 · PILOT SCORECARD

OUTCOME FAMILY	MEASURES
Consumer comprehension	Can explain payment, downside, trade-offs, residual risks and triggers in own words
Context quality	Relevant facts surfaced; contradictions resolved; missing information identified
Professional efficiency	Net discovery minutes; rework; follow-up contacts; exception handling
Decision quality	Material concerns resolved before commitment; alternatives considered; changes after clarification
Autonomy	Perceived pressure; ability to disagree; voluntary sharing; correction rate
Privacy / security	Optional disclosure rate; access-control failures; withdrawal requests; incidents
Overreliance	Acceptance of deliberately wrong or inconsistent system suggestions in test scenarios
Evidence quality	Source traceability; rationale completeness; reconstruction time; reviewer agreement

Source: 4orm research design.

Engagement is useful only if it serves these outcomes. A longer conversation that increases pressure or confusion is a failure. A shorter conversation that surfaces the one missing fact that changes the recommendation may be a success.

SECTION 42

Kill tests show what would disprove the model.

A credible research program needs release blockers that cannot be averaged away by a strong satisfaction score or a modest time saving.

EXHIBIT 39 · KILL TESTS

FAILURE CONDITION	RESPONSE
No meaningful comprehension improvement versus baseline	Do not claim decision-quality benefit; simplify or narrow the intervention
Consumers disclose materially more information than necessary or feel pressured to share	Redesign permission and collection model before scaling
Professionals find the handoff noisy or not decision-relevant	Reduce the output to a tighter decision brief; do not push more data
False reassurance or overreliance increases	Strengthen uncertainty language, role separation and human escalation
Performance is materially worse for lower-literacy, older, disabled or language-diverse users	Redesign accessibility and alternate channels before expansion
Better records reveal concerns but firms do not act on them	Do not position evidence capture as a complete control; require escalation workflow
Integration/review burden outweighs time saved or client benefit	Narrow the use case or stop deployment
A score begins replacing professional judgment	Remove or redesign the score; preserve the professional decision boundary

Source: 4orm research design.

> **A high satisfaction score should never offset a privacy failure. Faster workflow should never offset worse consumer understanding.**
>
> 4orm research principle

CONCLUSION

Prevent before the decision. Preserve evidence after it.

Industrial safety did not improve because organizations learned to investigate incidents more elegantly. The deeper change was to treat risk as something that should be identified, controlled and reassessed before exposure wherever possible. Field-level risk assessment, hierarchy of controls, process safety and Prevention through Design are different tools built around that same upstream logic. [1–9]

Financial services already have strong regulatory concepts - KYC, risk capacity, suitability, material-risk disclosure, privacy and consumer protection. The evidence gap is operational: meaningful context can still be incomplete, stale, emotionally difficult to disclose or poorly connected to the reasoning preserved in the final file. [10–14]

Counselling and conversational-AI research provides plausible mechanisms for a different front end: lower social pressure can sometimes increase candour; reflective and motivational approaches can preserve autonomy; conversational systems can increase engagement; structured intake can improve professional preparation; and longitudinal memory can preserve what mattered over time. The same evidence warns that poorly designed systems can feel judgmental, create over-trust or amplify error. [18–35]

The resulting 4orm thesis is therefore not “AI should give financial advice.” It is more restrained: give the person a private, consumer-controlled environment to understand the decision, identify future hazards, test controls and decide what to share; then give the professional better context and an evidence layer that shows what happened and why.

> **Industry learned to identify hazards before people went to work. 4orm can test whether the same preventive discipline belongs before people put their financial future to work.**
>
> Chad Johnston · Founder, 4orm

APPENDIX

Method, disclosure and evidence grading.

Method. This paper is a targeted narrative review completed in October 2026. It synthesizes occupational-safety guidance, Canadian regulatory materials, behavioural-finance and disclosure research, peer-reviewed studies of conversational systems, real-world deployments and company operating evidence. It also incorporates relevant findings and challenge disciplines from earlier 4orm research papers on financial decision quality and Financial Decision Intelligence.

EXHIBIT 40 · HOW THE EVIDENCE IN THIS PAPER IS GRADED

EVIDENCE LEVEL	USE IN THIS PAPER	TYPICAL LIMITATION
Government / regulator	Rules, guidance, safety methods, Canadian consumer context	May not establish product outcomes
Peer-reviewed experiments / meta-analyses	Mechanisms and causal or pooled effects	Population and domain transfer can be limited
Randomized trials	Comparative effects in a defined intervention	Product- and sector-specific
Large observational deployments	Workflow, access and engagement signals	Confounding and supplier involvement
Company operating metrics	Scale and implementation examples	Self-reported; selection and attribution risk
4orm analysis / founder perspective	Product hypothesis, architecture and questions	Requires direct validation

Source: 4orm method.

Disclosure

4orm is developing a commercial product in the area discussed and has an interest in the thesis. “Counselling-Centred AI,” “Future-Level Hazard Assessment,” “Financial Decision Intelligence” and the proposed architecture are 4orm concepts. None of the organizations cited has endorsed this paper or 4orm. Healthcare evidence is used as an adjacent mechanism or workflow analogue, not proof of financial outcomes. Supplier-reported metrics are identified as such.

Founder quotations are drawn from 4orm’s Founder Vision & North Star, Founder Charter and founder working-session statements, with light editing for readability where noted. The paper does not treat those statements as independent evidence.

SOURCE NOTES

Founder material, earlier research and boundaries.

Founder material used

4orm Founder Vision & North Star (October 2026); 4orm Founder Charter (October 2026); founder working-session statements from October 2026. Quotations marked “edited for clarity” preserve the meaning of the founder statement while removing spoken repetition.

Earlier 4orm research used

4orm Research Paper 01, “Financial decision quality and the evidence gap in Canada” (September 2026), contributed the Canadian evidence-gap framing, challenge discipline and pilot logic. 4orm Research Issue 02, “Financial Decision Intelligence” (October 2026), contributed the evidence grading, disclosure psychology, private-first decision-readiness thesis and earlier review of counselling and conversational-AI research. External claims were rechecked against current primary or original sources for this paper.

Terminology boundary

Field-Level Risk Assessment / Field-Level Hazard Assessment are established industrial terms used in different organizations. Future-Level Hazard Assessment is a 4orm concept and should always be identified as such. Counselling-Centred AI is also a 4orm design term: it describes a conversation philosophy and does not represent a licensed counselling service, diagnosis, treatment or substitute for regulated financial advice.

Final research position

> **Help the person understand the decision before it is made. Protect their privacy and autonomy. Give the professional better context. Preserve evidence of what happened and why.**
>
> 4orm permanent anchor

SOURCES

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Research paper · Issue 03, October 2026

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